5. *Design for failure*. Assume that sensors will fail, networks will be unreliable, and data will be missing. Build resilience into every layer of the architecture.

This architecture has proven robust in my home digital twin deployment, handling everything from brief network outages to sensor failures while maintaining continuous operation. The same principles scale to industrial deployments where the stakes are higher but the underlying challenges remain the same.

3.8 Summary

- Sensors are the bridge between the physical and digital worlds, and update a digital twin with changes detected in the real world.
- There are many types of sensors with different characteristics of resolution, sensitivity and accuracy and precision.
- Consider carefully what you need to measure, the performance you require and practical constraints such as where you must measure and what infrastructure you need when selecting sensors for your digital twin.
- Not every measurement that you need to take can be economically or practically captured by electronic sensors and manual sensing or integrating data that has been sensed by third parties is a critical aspect of capturing changes in the real world.
- Sensors need to communicate their readings at differing rates and distances and there are a variety of communication technologies to choose from.
- Sometimes you may not find the sensor that you require available off the shelf and will need to build your own.
- Operating sensors reliably at scale brings challenges related to battery management, firmware updates, calibration and physical asset tracking that can be addressed by automated health monitoring, standardization and documentation.